

Inflation Dynamics - Computational Notebook

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Abstract

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1. MEAN PROCESS SPECIFICATION

Four types of mean processes are of interest:

- **Constant:** $\pi_{t+1} = fc_t + \mu_{t+1}$
- **ARX(1,1):** $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 fc_t + \mu_{t+1}$
- **ARX(2,1):** $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 fc_t + \mu_{t+1}$
- **ARX(2,2)¹:** $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 fc_t + \phi_2 fc_{t-1} + \mu_{t+1}$

where μ_{t+1} is the residual term.

2. OLS RESULTS

I report OLS estimation results of three mean models with HAC standard error²:

ARX(1,1):

$$\begin{aligned}\hat{\pi}_{t+1} &= 0.0720 + 0.2881 \pi_t + 0.7508 fc_t \\ &= (0.087) \quad (0.114) \quad (0.168)\end{aligned}\tag{1}$$

ARX(2,1):

$$\begin{aligned}\hat{\pi}_{t+1} &= 0.0792 + 0.2793 \pi_t + 0.0728 \pi_{t-1} + 0.6661 fc_t \\ &= (0.087) \quad (0.102) \quad (0.127) \quad (0.257)\end{aligned}\tag{2}$$

ARX(2,2):

$$\begin{aligned}\hat{\pi}_{t+1} &= 0.0761 + 0.2880 \pi_t + 0.0799 \pi_{t-1} + 0.4720 fc_t + 0.1793 fc_{t-1} \\ &= (0.088) \quad (0.109) \quad (0.127) \quad (0.440) \quad (0.295)\end{aligned}\tag{3}$$

¹In the GARCH family estimation exercise, we found that ARX(2,2) is always dominated by other three specifications. Thus, we stop estimating ARX(2,2) in BEGE estimation to saving computational resources.

²Standard errors are heteroskedasticity and autocorrelation robust (HAC) using 12 lags and without small sample correction.

GARCH model specification:

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{k=1}^q \beta_k \sigma_{t-k}^2 \quad (4)$$

3. GJR-GARCH SPECIFICATION

GJR GARCH model specification:

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i |u_{t-i}|^2 + \sum_{j=1}^o \gamma_j |u_{t-j}|^2 I[u_{t-j} < 0] + \sum_{k=1}^q \beta_k \sigma_{t-k}^2 \quad (5)$$

4. INFLATION DYNAMICS

To start the random search of initial mean parameters, I draw uniform samples from $(\mu - 2\sigma, \mu + 2\sigma)$ where μ and σ are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

Table 1: Parameter Bounds for Mean Process Specifications

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	—	—	—	—	—
ARX(1,1)	$(\min \pi_t, \max \pi_t)$	$(-0.999, 0.999)$	—	$(-10, 10)$	—
ARX(2,1)	$(\min \pi_t, \max \pi_t)$	$(-1.999, 1.999)$	$(-0.999, 0.999)$	$(-10, 10)$	—
ARX(2,2)	$(\min \pi_t, \max \pi_t)$	$(-1.999, 1.999)$	$(-0.999, 0.999)$	$(-10, 10)$	$(-10, 10)$

4.a. II. GARCH Model Summarization

Table 2: Model selection by AIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	359.4710	(1,1)	(1,1)	368.2429
t	(1,1)	(2,1)	343.1431	(2,1)	(1,1)	344.6254
Skew t	(1,1)	(2,1)	345.1312	(2,1)	(1,1)	346.5791
GED	(1,1)	(2,1)	345.1925	(1,1)	(1,1)	347.6934
Mix of Normal	(1,1)	(2,1)	345.8222	(2,1)	(2,1)	343.5023

Table 3: Model selection by BIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	381.4280	FC	(1,1)	383.4754
t	FC	(1,1)	367.8350	FC	(1,1)	367.1732
Skew t	FC	(1,1)	370.4242	FC	(1,1)	369.6649
GED	FC	(1,1)	371.8489	FC	(1,1)	370.9644
Mix of Normal	FC	(1,1)	373.7106	FC	(1,1)	371.0689

EGARCH model specification:

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \left(|e_{t-i}| - \sqrt{2/\pi} \right) + \sum_{j=1}^o \gamma_j e_{t-j} + \sum_{k=1}^q \beta_k \ln \sigma_{t-k}^2 \quad (6)$$

where $e_t = \varepsilon_t / \sigma_t$

5. BEGE DENSITY

BEGE density function $f_{BEGE}(\mu | p, n, \sigma_p, \sigma_n)$ is the function that calculates the density of the observation μ given the parameters $\{p, n, \sigma_p, \sigma_n\}$.

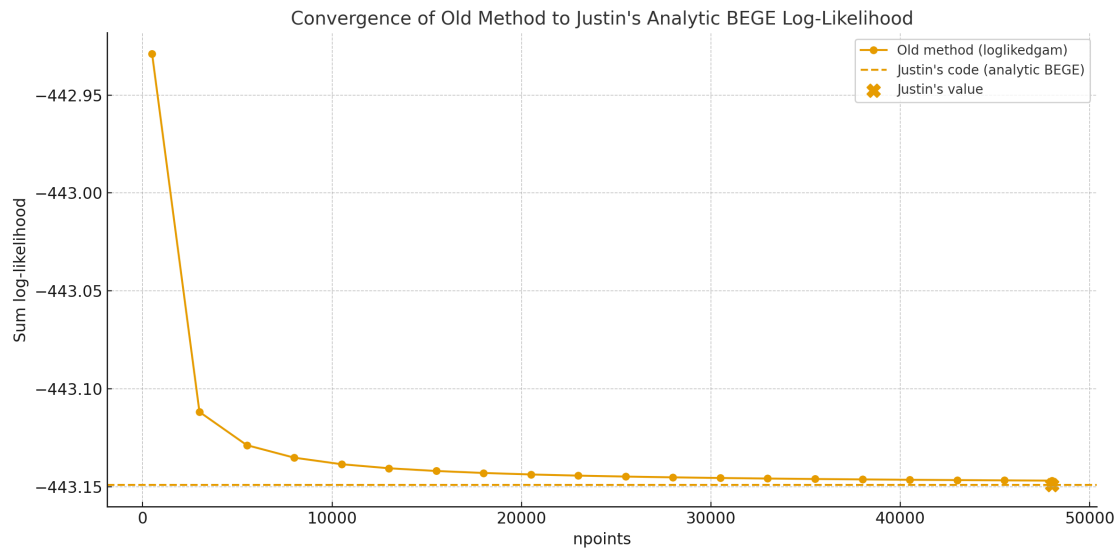
To compare Justin's analytic BEGE density code with the old numerical one, I conducted an experiment using synthetic data. I generated 200 independent observations from a standard normal distribution and fixed the BEGE parameters to

$$p = 5, \quad n = 7, \quad \sigma_p = 0.8, \quad \sigma_n = 1.2. \quad (7)$$

I computed the sum of log-likelihood using two approaches:

1. The *old BEGE function*, which approximates the density via numerical integration on a uniform grid, and whose accuracy depends strongly on the number of grid points; and
2. *Justin's analytic BEGE function*.

The old numerical method was evaluated over a wide range of grid resolutions (up to 50,000 points), and the resulting log-likelihood sums were plotted alongside the benchmark value computed from Justin's code.



Findings:

- The old numerical likelihood converges toward Justin's analytic likelihood as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood. This explains why, for a given set of parameters, Justin's implementation produces a lower log-likelihood than the old code.
- Justin's analytic BEGE formula provides a more accurate and more computationally efficient evaluation.

6. BEGE GARCH FAMILY

6.a. Constant BEGE

The first model that I estimated is BEGE with invariant shape parameters, which is

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n\end{aligned}\tag{8}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1).\tag{9}$$

$\bar{p}$ and $\bar{n}$ are unconditional shape parameters. The random search and bounds are set to be:

- $0.05 < \sigma_p, \sigma_n < 2$
- $0.1 < p, n < 10$

6.b. Symmetric BEGE

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned}\pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n,\end{aligned}\tag{10}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1).\tag{11}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho, \phi^+, \phi^-)$:

$$\begin{aligned}p_t &= p_0 + \rho p_{t-1} + \frac{\phi^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho n_{t-1} + \frac{\phi^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi^-}{2\sigma_n^2} (u_{t-1}^-)^2\end{aligned}\tag{12}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho < 0.999$,
- $10^{-5} < \phi^+, \phi^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

6.c. Bad and Good Symmetric GARCH Model

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n)$:

$$\begin{aligned}p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p}{2\sigma_p^2} (u_{t-1})^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n}{2\sigma_n^2} (u_{t-1})^2\end{aligned}\tag{13}$$

I have set the constraints below.

- $\rho + \phi < 1$ for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$.

- $\max\{p_t, n_t\} < 200$

6.d. Inflation and Deflation Model

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2 \end{aligned} \quad (14)$$

According to the emails, I have set the constraints below.

- $\rho_p + \frac{\phi_p^+}{2} < 1$ and $\rho_n + \frac{\phi_n^-}{2} < 1$.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$.
- $\max\{p_t, n_t\} < 200$

6.e. Full BEGE

BEGE GARCH has mean process and higher:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n, \end{aligned} \quad (15)$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \quad (16)$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} (u_{t-1}^+)^2 + \frac{\phi_p^-}{2\sigma_p^2} (u_{t-1}^-)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} (u_{t-1}^+)^2 + \frac{\phi_n^-}{2\sigma_n^2} (u_{t-1}^-)^2 \end{aligned} \quad (17)$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho_p, \rho_n < 0.999$,
- $10^{-5} < \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$ for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \text{Var}(\pi_t)$ where $\text{Var}(\pi_t) = 0.75$ is the unconditional variance of inflation.
- $\max\{p_t, n_t\} < 200$.

7. RESULT SUMMARY

Overall, most of the estimations with random initial values converge. I randomly sampled 100,000 starting values for the constant BEGE model, and sampled 10,000 for symmetric BEGE because it is more time consuming. I report the best estimation below.